Report Title

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1 Introduction

1.1 Ordered External Electric Fields as catalysts

Chemical reactivity can be broadly reduced down to the dynamics of electrons, it stands to reason that being able to utilise electrostatics to manipulate the electrons that govern reactivity is a grand challenge of chemistry. This notion has been at the forefront of chemical research for as long as electron directing groups have been utilised to manipulate the strength of bonds, however the idea of influencing a reaction from a distance using electrostatics has only been explored in the past century.

The methods to this approach were initially quite crude, involving the use of homogenous ionic species,[1] though through the use of early *ab initio* techniques, the methods were able to be more rigorously and precisely investigated. These studies in the early 90s sought to understand solvent effects and more generally the effects of, localised electric fields (EF) on chemical reactions, such as the Friedel-Crafts[2] and Menshutkin[3] reactions. Further evidence for the direct influence of electrostatic catalysis came from biology, as the idea that within an enzyme, an internal electric field (IEF) generated by charged amino acids within the binding site could effectively be creating a pre-organised environment that would stabilise a substrate's transition state (TS).[4]

Whilst developing the valence bond model of chemical reactivity, Shaik[5] recognised that the definitions of chemical bonding that emerged from the theory indicated that all bonds, irrespective of the nature of their component atoms could be described in a polar context. This inspired Sini et al.[6] to explore the idea, developing a model of bonding that described all bonds as a superposition of a spin paired, purely covalent interaction and two possible ionic interactions,

as shown in Equation (1).[7]

$$\Psi_{A-B} = C_1(\mathrm{A{-}B}) + C_2(\mathrm{A^- B^+}) + C_3(\mathrm{A^+B^-}) \tag{1}$$

These theoretical formalisms laid the groundwork for further computational research into the field, with Shaik et al. continuing to explore the effects of external electric fields (EEF) to first, create selectivity between competing nonpolar epoxidation and hydroxylation reactions[8] and subsequently for Meir et al.[9] to identify the endo/exo selectivity of Diels-Alder reactions, afforded by the use of oriented EEFs (OEEFs). This theoretical exploration was put to the test experimentally by Aragonès et al.,[10] who used an OEFF directed from a scanning tunnelling microscope plate to the tip, to not only control the endo/exo selectivity of the reaction, but also to quantify and correlate the effect of the EF strength.

With sufficient experimental evidence and theoretical understanding of the effect of OEFFs, Shaik et al. formalised the "governing effects and rules for electric-field effects on chemical reacivity",[7] briefly summarised as follows:

(a) An OEEF that is oriented along a "bond axis" will polarise the bond, favouring one of the ionic species (Equation (1)), eventually cleaving it heterolytically. As this polarisation occurs, the bond dipole increases, causing it to interact more strongly with the OEEF, stabilising the species in the process.

(b) Every reaction possesses a "reaction axis" which follows the flow of electrons from reactant to product, through the TS.

(c) Applying an OEEF along the "reaction axis" will stabilise the TS by increasing the bond ionicity (as per rule (a)) and, symmetry of the HOMO-LUMO overlap of the transition species allowing, will encourage mixing of the charge transfer states.

(d) A molecule with a dipole moment or that is polarisable, will rotate to align with a sufficiently strong OEFF.

1.2 Scaling OEEFs to Bulk Catalysis

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- Marcus theory and reducing reorganisation energy - Landau[11]

– Breaking the solvation shell - Novák et al.[12]

- Restructuring of the solvent with an EEF - Daniels et al.[13]

- EEF "tweezing" effect/aligning of reacting species (Shaik rule 4) - Wang et al.,[14] Shaik et al.[7]

$$\Delta E = 4.8\vec{F} \cdot \vec{\mu}\,\mathrm{kcal{\cdot}mol^{-1}} \tag{2}$$

- Screening of the EEF by the solvent - Xu et al.,[15] Shida et al.[16]

- Using pulsed EEFs to reduce electrochemical reactivity - Xu et al.,[15] Shaik et al.[7]

- Specialist equipment required to mitigate polar solvent screening - Xu et al.,[15] Shaik et al.,[7] Shida et al.[16]

1.3 Creating IEFs from Ordered Ionic Liquids

- ILs - MacFarlane and Seddon[17]

- PILs - Greaves and Drummond[18]

- polarisability (leads to the IEF) need ref - Brehm et al.[19] (delocalisation of e-density)

- ILs slow migration makes them good candidates for long lasting IEF bearers with pulsed EEFs need ref

- ILs ordering in EEF - Zhao et al., Wang[20,21]

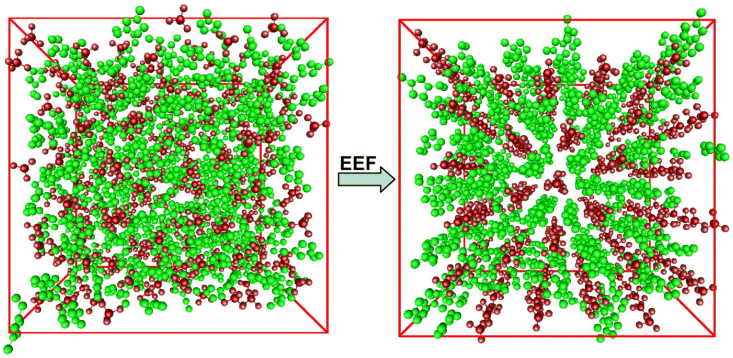

Figure 1.1: Snapshot from a molecular dynamics simulation of [C_2mim][BF_4] without (left) and with an EEF of $1.14\,\mathrm{V{\cdot}\AA^{-1}}$ (right). C_2mim is coloured green and BF_4 is coloured red, Sourced from Zhao et al..[20]

- proof of concept for ILs - Xu et al.[15]

- Adding h-bonding - [$c_2$mim][OAc] in equilibrium with carbene-acetate form - Brehm et al.[19]

re-read

–H-bonding network in ILs - Canongia Lopes and Pádua[22]

– Ethyl chain to reduce the amount of nonpolar aggregation that would disrupt the H-bonding network - Canongia Lopes and Pádua[22]

– Notion of pK_a is broken in pure IL solutions, as it's an aqueous property, proton transfer happens "orders of magnitude" less - MacFarlane and Seddon[17]

- Power of IL intramolecular interactions

– Dissolution of cellulose? - Swatloski et al.[23] read this article Mohd et al.[24] and this one

1.4 Computational Chemistry

Just wing this section? you know this bit to death... - QM - Cramer[25]

- DFT - Cramer[25]

– empirical dispersion correction for accurate polarisation - Izgorodina et al.[26]

- MD

– since large scale calcs aren't viable

– pMD to account for dispersion effects - Bedrov et al.[27]

1.5 Research Objectives

2 References

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